

Homework 11 – Solutions

8.6 (d) Since $X \sim \text{Geom}(p)$ and $Y \sim \text{Binom}(n, r)$, we know that

$$E[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}, \quad E[Y] = nr, \quad \text{Var}(Y) = nr(1-r).$$

From these we can compute the second moments:

$$E[X^2] = \text{Var}(X) + (E[X])^2 = \frac{2-p}{p^2}, \quad E[Y^2] = \text{Var}(Y) + (E[Y])^2 = n(n-1)r^2 + nr.$$

To compute $E[(X+Y)^2]$ we first expand the square and use linearity:

$$E[(X+Y)^2] = E[X^2 + 2XY + Y^2] = E[X^2] + 2E[XY] + E[Y^2].$$

We already have $E[X^2]$ and $E[Y^2]$. For $E[XY]$ we use that these random variables are independent and hence

$$E[XY] = E[X]E[Y].$$

This gives

$$\begin{aligned} E[(X+Y)^2] &= E[X^2] + 2E[XY] + E[Y^2] \\ &= \frac{2-p}{p^2} + \frac{2nr}{p} + n(n-1)r^2 + nr. \end{aligned}$$

Another approach: Since the random variables are independent we have $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$. This gives

$$\begin{aligned} E[(X+Y)^2] &= \text{Var}(X+Y) + (E[X+Y])^2 = \text{Var}(X) + \text{Var}(Y) + (E[X] + E[Y])^2 \\ &= \frac{1-p}{p^2} + nr(1-r) + (1/p + nr)^2, \end{aligned}$$

which (after some algebra) leads to the same expression as the previous solution.

8.12 (a) We need to compute

$$M_Z(t) = E(e^{tZ}) = \int_{-\infty}^{\infty} e^{tz} f_Z(z) dz = \int_0^{\infty} e^{tz} \lambda^2 z e^{-\lambda z} dz = \lambda^2 \int_0^{\infty} z e^{-(\lambda-t)z} dz.$$

If $\lambda - t \leq 0$ then this integral is at least as large as $\lambda^2 \int_0^{\infty} z dz$ which is infinite. If $\lambda - t > 0$ then we can compute the integral using integration by parts to give

$$M_Z(t) = \begin{cases} \frac{\lambda^2}{(\lambda-t)^2}, & \text{if } t < \lambda \\ \infty, & \text{if } t \geq \lambda. \end{cases}$$

(b) We have already computed the MGF of an exponential distribution (see e.g. Example 5.6), so

$$M_X(t) = M_Y(t) = \begin{cases} \frac{\lambda}{\lambda-t}, & \text{if } t < \lambda \\ \infty, & \text{if } t \geq \lambda. \end{cases}$$

Since X and Y are independent, we have $M_{X+Y}(t) = M_X(t)M_Y(t)$. Comparing with part (a) we see that $X+Y$ has the same moment generating function as Z , which means that they must have the same distribution. (Since the moment generating function is finite in a neighborhood of 0.)

8.22 There are several ways to approach this problem. One possibility that gives the answer without doing complicated computations is as follows. For each $1 \leq j \leq 89$ let I_j be the indicator of the event that both j and $j + 1$ are chosen among the five numbers. Then $X = \sum_{j=1}^{89} I_j$, since if j and $j + 1$ are both chosen then they will be next to each other in the ordered sample. By linearity

$$E[X] = E\left[\sum_{j=1}^{89} I_j\right] = \sum_{j=1}^{89} E[I_j].$$

We can compute $E[I_j]$ directly by counting favorable outcomes:

$$E[I_j] = P(\text{both } j \text{ and } j + 1 \text{ are chosen}) = \frac{\binom{88}{3}}{\binom{90}{5}} = \frac{2}{801}.$$

Thus

$$E[X] = 89 \cdot \frac{2}{90 \times 89} = \frac{20}{90} = \frac{2}{9}.$$

Here is another possible solution: for any $1 \leq a < b \leq 5$ let $J_{a,b}$ be the indicator that the a th and b th picks are next to each other. (Note that the original 5 numbers are chosen with order!) Then $X = \sum_{1 \leq a < b \leq 5} J_{a,b}$ and because of the exchangeability of the 5 picks we have $E[J_{1,2}] = E[J_{a,b}]$ for any $1 \leq a < b \leq 5$. Hence by the linearity of expectation we have

$$E[X] = \sum_{1 \leq a < b \leq 5} E[J_{a,b}] = \binom{5}{2} E[J_{1,2}]$$

We have

$$E[J_{1,2}] = P(\text{the first two picks are next to each other}).$$

This probability can be computed by just considering the first two picks (and disregarding order) to get

$$E[J_{1,2}] = \frac{89}{\binom{90}{2}} = \frac{2}{90} = \frac{1}{45}$$

(since out of the equally likely $\binom{90}{2}$ pairs there are exactly 89 where we have two neighboring numbers). This gives

$$E[X] = \binom{5}{2} \frac{1}{45} = \frac{10}{45} = \frac{2}{9}.$$

8.23 (a) For $i = 1, \dots, 50$, let

$$Y_i = \begin{cases} 0, & \text{the } i\text{th pick is red;} \\ 1, & \text{the } i\text{th pick is green} \end{cases}$$

Then $Y_1, \dots, Y_{50}$ are exchangeable so

$$\begin{aligned} P(Y_{28} \neq Y_{29}) &= P(Y_{28} = 0, Y_{29} = 1) + P(Y_{29} = 0, Y_{28} = 1) \\ &= 2P(Y_1 = 0, Y_2 = 1) = 2 \left(\frac{20 \cdot 30}{50 \cdot 49} \right) = \frac{24}{49}. \end{aligned}$$

(b) Let I_j be the indicator that $Y_j \neq Y_{j+1}$ for $j = 1, \dots, 49$. Then $X = I_1 + \dots + I_{49}$ and by linearity

$$E[X] = \sum_{i=1}^{49} E[I_i] = \sum_{i=1}^{49} P(Y_i \neq Y_{i+1})$$

By the exchangeability of the Y_i random variables and part (a) we get

$$E[X] = \sum_{i=1}^{49} P(Y_i \neq Y_{i+1}) = 49P(Y_1 \neq Y_2) = 49 \left(\frac{24}{49} \right) = 24.$$

8.36 (a) Let I_k be the indicator of the event that the number k appears at least once among the four die rolls. Then $X = I_1 + \dots + I_6$ and we get

$$E[X] = E[I_1 + \dots + I_6] = E[I_1] + \dots + E[I_6] = 6E[I_1],$$

where the last step comes from exchangeability. We have

$$E[I_1] = P(\text{the number 1 shows up}) = 1 - P(\text{none of the rolls are equal to 1}) = 1 - \left(\frac{5}{6} \right)^4$$

which gives

$$E[X] = 6 \left(1 - \left(\frac{5}{6} \right)^4 \right).$$

(b) We can compute the variance either by computing $E[X^2]$ (and using part (a)) or using the formula for the variance of the sum. We will use the second approach.

We have $X = I_1 + \dots + I_6$, so

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^6 I_i\right) = \sum_{i=1}^6 \sum_{j=1}^6 \text{Cov}(I_i, I_j).$$

Considering the $i = j$ and $i \neq j$ terms separately:

$$\text{Var}(X) = \sum_{i=1}^6 \text{Var}(I_i) + \sum_{i \neq j} \text{Cov}(I_i, I_j)$$

By exchangeability $\text{Var}(I_i) = \text{Var}(I_1)$ for all i , and $\text{Cov}(I_i, I_j) = \text{Cov}(I_1, I_2)$ for all $i \neq j$. Counting the respective terms gives

$$\text{Var}(X) = 6 \text{Var}(I_1) + 30 \text{Cov}(I_1, I_2).$$

The variance of a Bernoulli(p) random variable is $p(1 - p)$, so using part (a) we get

$$\text{Var}(I_1) = \left(1 - \left(\frac{5}{6} \right)^4 \right) \left(\frac{5}{6} \right)^4.$$

Recall that for two indicator random variables I_A, I_B (where A and B are the respective events) we have

$$\text{Cov}(I_A, I_B) = P(AB) - P(A)P(B).$$

To compute $\text{Cov}(I_1, I_2)$ the corresponding events are

$$A = \{\text{the number 1 shows up}\}, \quad B = \{\text{the number 2 shows up}\}.$$

We have seen in (a) that $P(A) = P(B) = \left(1 - \left(\frac{5}{6}\right)^4\right)$. To compute $P(AB)$ we consider the complement $P((AB)^c) = P(A^c \cup B^c)$ and use inclusion-exclusion:

$$P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c B^c) = \left(\frac{5}{6}\right)^4 + \left(\frac{5}{6}\right)^4 - \left(\frac{2}{3}\right)^4$$

where we used the fact that $A^c B^c$ is the event that neither 1 or 2 shows up among the numbers. This gives

$$P(AB) = 1 - 2\left(\frac{5}{6}\right)^4 + \left(\frac{2}{3}\right)^4$$

and

$$\text{Cov}(I_1, I_2) = 1 - 2\left(\frac{5}{6}\right)^4 + \left(\frac{2}{3}\right)^4 - \left(1 - \left(\frac{5}{6}\right)^4\right)^2 = \left(\frac{2}{3}\right)^4 - \left(\frac{5}{6}\right)^8$$

Collecting everything gives

$$\text{Var}(X) = 6\left(1 - \left(\frac{5}{6}\right)^4\right)\left(\frac{5}{6}\right)^4 + 30\left(\left(\frac{2}{3}\right)^4 - \left(\frac{5}{6}\right)^8\right) \approx 0.447.$$

Alternate solution if you want to compute $\text{Var}(X)$ via $E[X^2]$:

$$\begin{aligned} E[X^2] &= E[(I_1 + \cdots + I_6)^2] = E\left[\sum_{k=1}^6 I_k^2 + 2 \sum_{j < k \leq 6} I_j I_k\right] \\ &= \sum_{k=1}^6 E[I_k^2] + 2 \sum_{j < k \leq 6} E[I_j I_k]. \end{aligned}$$

Since I_k is either 0 or 1, we have $I_k^2 = I_k$. Using exchangeability

$$\begin{aligned} E[X^2] &= \sum_{k=1}^6 E[I_k^2] + 2 \sum_{j < k \leq 6} E[I_j I_k] \\ &= \sum_{k=1}^6 E[I_k] + 2 \sum_{j < k \leq 6} E[I_j I_k] \\ &= 6E[I_1] + 30E[I_1 I_2]. \end{aligned}$$

From this point one can proceed as in the previous solution to get $E[I_1]$ (from part (a)) and $E[I_1 I_2]$ (as the probability $P(AB)$ above). This leads to the same answer:

$$E[X^2] = 6\left(1 - \left(\frac{5}{6}\right)^4\right) + 30\left(1 + \left(\frac{2}{3}\right)^4 - 2 \cdot \left(\frac{5}{6}\right)^4\right)$$

and

$$\begin{aligned} \text{Var}(X) &= E[X^2] - E[X]^2 \\ &= 6\left(1 - \left(\frac{5}{6}\right)^4\right) + 30\left(1 + \left(\frac{2}{3}\right)^4 - 2 \cdot \left(\frac{5}{6}\right)^4\right) - 36\left(1 - \left(\frac{5}{6}\right)^4\right)^2 \\ &\approx 0.447. \end{aligned}$$

8.40 (a) Let J_k be the indicator for the event that the number k is drawn in at least one of the 4 weeks. Then $X = J_1 + J_2 + \cdots + J_{90}$. Then by the linearity of expectation and exchangeability we get

$$E[X] = E\left[\sum_{k=1}^{90} J_k\right] = \sum_{k=1}^{90} E[J_k] = 90E[J_1].$$

We have

$$\begin{aligned}
E[J_1] &= P(1 \text{ is drawn at least one of the 4 weeks}) \\
&= 1 - P(1 \text{ is not drawn in any of the 4 weeks}) \\
&= 1 - \left(\frac{89 \cdot 88 \cdot 87 \cdot 86 \cdot 85}{90 \cdot 89 \cdot 88 \cdot 87 \cdot 86} \right)^4 = 1 - \left(\frac{85}{90} \right)^4.
\end{aligned}$$

From this

$$E[X] = 90E[J_1] = 90 \left(1 - \left(\frac{85}{90} \right)^4 \right) \approx 18.394.$$

8.42 We have

$$E[\bar{X}_n^4] = E \left[\left(\frac{X_1 + \cdots + X_n}{n} \right)^4 \right] = \frac{1}{n^4} E[(X_1 + \cdots + X_n)^4].$$

By expanding the fourth power of the sum and using linearity and exchangeability

$$\begin{aligned}
E[\bar{X}_n^4] &= \frac{1}{n^4} E \left[\sum_{k=1}^n X_k^4 + 24 \sum_{i < j < k < \ell} X_i X_j X_k X_\ell \right. \\
&\quad \left. + 12 \sum_{\substack{k < \ell \\ j \neq k, j \neq \ell}} X_j^2 X_k X_\ell + 6 \sum_{j < k} X_j^2 X_k^2 + 4 \sum_{j \neq k} X_j^3 X_k \right] \\
&= \frac{1}{n^4} \sum_{k=1}^n E[X_k^4] + 24 \sum_{i < j < k < \ell} E[X_i X_j X_k X_\ell] \\
&\quad + 12 \sum_{\substack{k < \ell \\ j \neq k, j \neq \ell}} E[X_j^2 X_k X_\ell] + 6 \sum_{j < k} E[X_j^2 X_k^2] + 4 \sum_{j \neq k} E[X_j^3 X_k] \\
&= \frac{1}{n^3} E[X_1^4] + 24 \binom{n}{4} E[X_1 X_2 X_3 X_4] \\
&\quad + 12 \cdot \binom{n}{3} E[X_1^2 X_2 X_3] + 6 \binom{n}{2} E[X_1^2 X_2^2] + 4n(n-1) E[X_1^3 X_2].
\end{aligned}$$

By independence

$$\begin{aligned}
E[X_1 X_2 X_3 X_4] &= E[X_1] E[X_2] E[X_3] E[X_4] = 0, & E[X_1^2 X_2 X_3] &= E[X_1^2] E[X_2] E[X_3] = 0, \\
E[X_1^3 X_2] &= E[X_1^3] E[X_2] = 0, & E[X_1^2 X_2^2] &= E[X_1^2] E[X_2^2] = E[X_1^2]^2 = a^2.
\end{aligned}$$

Hence

$$E[\bar{X}_n^4] = \frac{1}{n^3} E[X_1^4] + \frac{3n(n-1)}{n^4} E[X_1^2]^2 = \frac{c}{n^3} + \frac{3(n-1)a^2}{n^3}.$$

8.48 The joint probability mass function of the random variables (X, Y) can be represented by the

following table.

		Y		
		0	1	2
X	1	$\frac{9}{100}$	0	0
	2	$\frac{81}{100}$	$\frac{9}{100}$	0
	3	0	0	$\frac{1}{100}$

Hence, the marginal distribution are:

$$\begin{aligned} p_X(1) &= \frac{9}{100}, & p_X(2) &= \frac{90}{100}, & p_X(3) &= \frac{1}{100} \\ p_Y(0) &= \frac{90}{100}, & p_Y(1) &= \frac{9}{100}, & p_Y(2) &= \frac{1}{100}. \end{aligned}$$

From these we can compute the following expectations:

$$E[X] = \frac{48}{25}, \quad E[Y] = \frac{11}{100}, \quad E[XY] = \frac{6}{25},$$

and so

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = \frac{6}{25} - \frac{48}{25} \cdot \frac{11}{100} = \frac{18}{625}.$$

8.54 (a) We have

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 = 5 - 2^2 = 1 \\ \text{Var}(Y) &= E[Y^2] - (E[Y])^2 = 10 - 1^2 = 9 \\ \text{Cov}(X, Y) &= E[XY] - E[X]E[Y] = 1 - 2 \cdot 1 = -1. \end{aligned}$$

Then

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{-1}{\sqrt{1 \cdot 9}} = -\frac{1}{3}.$$

(b) We have

$$\text{Cov}(X, X + cY) = \text{Var}(X) + c \text{Cov}(X, Y) = 1 + c(-1) = 1 - c.$$

Thus for $c = 1$ the random variables X and $X + cY$ are uncorrelated.